L.J Institute of Engineering and Technology

Data Structures using JAVA PRACTICE BOOK (SEM-II EVEN_2025-CE RELATED BRANCHES)

Note :

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Date : 10 Mar 2025

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
1	1	User defined data type is also called?	B	0.5	Primitive	Non-primitive	Identifier	None of these
2	1	Non Primitive data Structures are those which define set of _____.	B	0.5	Static Elements	Derived Elements	Primitive Data	None of these
3	1	Which of the following is not a primitive data structure?	C	0.5	Boolean	Integer	Arrays	Character
4	1	Which of the following is a primitive data structure?	B	0.5	Stack	Integer	Arrays	Queue
5	1	The way in which the data item or items are logically related defines……	B	0.5	storage structure	data structure	data Relationship	data operation
6	1	Float is a _____ data type.	A	0.5	Primitive	Non Primitive	Linear	Non Linear
7	1	The data structure used in hierachical data model is	D	0.5	Array	Stack	Graph	Tree
8	1	If elements of the data structure forms a sequence of list then it is called as _____.	A	0.5	Linear data structure	Primitive data structure	Non-primitive data structure	None of these
9	1	Which of the following is a linear data structure?	A	0.5	Array	AVL trees	Binary Trees	Graphs
10	1	The data structure which is active only at one end is……..	B	0.5	queue	stack	linked list	tree
11	1	Which of the following is a non linear data structure?	D	0.5	Array	Linked list	Stack	Tree
12	1	Which of the following is the disadvantage of the array?	C	0.5	Stack and Queue data structures can be implemented through an array.	Index of the first element in an array can be negative	Wastage of memory if the elements inserted in an array are lesser than the allocated size	Elements can be accessed sequentially.
13	1	In which Data Structure, Insertion and Deletion can be performed at the one end?	C	0.5	queue	Linkedlist	Stack	Tree
14	1	In given array A [-3…5] [-1…3], what is the value of Upper Bound of Row Index?	B	0.5	-3	5	8	-1
15	1	In Column major, how is the following matrix stored in memory? 32 36 35 31 34 33 37 39 38	A	1	32 31 37 36 34 39 35 33 38	39 35 33 38 32 31 37 36 34	37 36 34 39 35 33 38 32 31	33 37 39 38 32 36 35 31 34
16	1	Which of the following is the advantage of the array data structure?	B	1	Elements of mixed data types can be stored.	Easier to access the elements in an array	Index of the first element starts from 1.	Elements of an array cannot be sorted
17	1	Which of the following data structure can't store the non-homogeneous data elements?	A	1	Arrays	Records	Objects	None of the above
18	1	A Data structure which displays the relationship of adjacency between elements is said to be	A	1	Linear data structure	Trees	Non Linear data structure	None of these

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19	1	Which of the following are linear type of data structure? i) Linked list ii) Stack iii) Binary Tree iv) Array v) Queue	C	1	i, ii, iii and iv only	ii, iii, iv and v only	i, ii, iv and v only	All i, ii, iii, iv and v
20	1	Match the following properties of an array with their descriptions. a) Homogeneous i) the list size is constant. b) Ordered ii) there is a first and last element. c) Finite iii) there is a next and previous in the natural order of the structure d) fixed-length iv) every element is the same type.	D	1	a-i, b-ii, c-iii, d-iv	a-ii, b-iii, c-iv, d-i	a-iii, b-i, c-ii, d-iii	a-iv, b-iii, c-ii, d-i
21	1	The elements of a linked list are stored	C	0.5	In a structure	In an array	Anywhere the computer has space for them	In contiguous memory locations
22	1	Linear arrays are also called…	A	0.5	One-dimensional array	Vertical Array	Horizontal Array	All of the above
23	1	If more than one subscript is used, an array is known as a_____.	C	0.5	One- dimensional array	Single dimensional array	Multi- dimensional array	None of the above
24	1	Each element in an array is associated with a unique subscript value, starting from_____.	B	0.5	1 to size-1	0 to size-1	-1 to size-0	None of the above
25	1	In a 4*4 matrix if the address of A[1][1] and A[2][1] are 1024 and 1040 respectively and each element occupies 4 bytes then the array has been stored in _____ and the number of elements in between the elements A[1][1] and A[2][1] is _____.	A	1	RMO, 3	CMO, 5	RMO, 4	CMO, 4
26	1	In a 6*6 matrix if the address of A[3][2] and A[3][3] are 1024 and 1036 respectively and each element occupies 2 bytes then the array has been stored in _____ and the number of elements in between the elements A[3][2] and A[3][3] is……	C	1	RMO, 5	RMO, 6	CMO, 5	CMO, 6
27	1	Given array A[3][4], base address=100, storage size=2 bytes. Find the location of A[2][1], if array is stored in column wise?	D	1	118	120	115	110
28	1	Array elements are stored in	A,B	1	row major	column major	major matrix	minor row
29	1	Data elements grouped in an array can be of any basic data type like:	D	1	integer	float	double	All of the above
30	1	Given array A[9], base address 150, storage size = 1 byte, Find the location of A[3].	B	1	156	153	159	154

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31	1	Let 'A' be a two-dimensional array declares as follows: A: array [1….15] [1……20] of integer type and size of each element is one byte, The array is stored in column major order and first element of array is stored at location 150, What is the address of the element A[i][j]?	B	1	15i+j+84	15j+i+134	10i+j+89	10j+i+89
32	1	Given array A[0..2][0..2], base address=1000, storage size=1 byte. Find the location of A[1][1], if array is stored in row wise?	A	1	1004	1008	1012	1010
33	1	The largest subscript of an array index is called its	C	1	lower bound	range	upper bound	All of these
34	1	The given array is P[-4…6][-2…10] is a integer type array. An elements are stored in ROW MAJOR order. Calculate how many no of columns are in this array, if base address is 70.	A	1	13	11	12	15
35	1	The Smallest subscript of an array index is called its	A	1	lower bound	range	upper bound	All of these
36	1	A programmer can access a particular element of an array by using one or more_____.	D	1	Indices	Subscripts	Single-dimensional array	Both (a) and (b)
37	1	If row-major order is used, how is the following matrix stored in memory? a b c d e f g h i	B	1	ihgfedcba	abcdefghi	cfibehadg	adgbehcfi
38	1	If column-major order is used, how is the following matrix stored in memory? a b c d e f g h i	D	1	ihgfedcba	abcdefghi	cfibehadg	adgbehcfi
39	1	Row -major order in two -dimentional array refers to an arrangement where	A	1	all elements of a row are stored in memory in sequence followed by next row in sequence,and so on	all elements of row are stored in memory in sequence followed by next column in sequence ,and so on	all elements of column are stored in memory in sequence followed by next column in sequence, and so on	none of the above
40	1	Given array A[3][4], base address=100, storage size=2 bytes. Find the location of A[2][1], if array is stored in Row wise?	A	1	118	120	115	110
41	1	Calculate the address of M[-3]. Where given array is M[-9,7] with base address 1002. Consider the size of single element as 4Bytes.	C	1	1004	1020	1026	1030

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42	1	10 11 13 12 14 17 15 16 18 If column major order given then the sequence of element stored in memory.	A	1	10 12 15 11 14 16 13 17 18	10 11 15 12 14 16 13 17 18	10 11 12 15 14 16 13 17 18	10 11 14 11 15 16 13 17 18
43	1	Given array is A [4…7] [-1…3] and storage size = 2 with BA= 100 and calculate the address of A [6] [2] an element stored in column major order.	A	1	128	125	126	127
44	1	If given array is A[-57---- 42] with base address 1050 & array is integer type of array. Calculate the total no of elements in Array.	A	1	100	102	99	98
45	1	Consider an array A[20,10], assume 2B per memory cell and the base address of array is 100. What is the address of [11,5]? Assume Column major storage.	C	1	330	320	322	None
46	1	If the address of A[2][2] and A[3][2] are 2020 and 2024 respectively and each element occupies 4 bytes then the array has been stored in _____ order.	B	1	row major	column major	major matrix	minor row
47	1	In a 4*4 matrix if the address of A[1][1] and A[2][1] are 1000 and 1008 respectively and each element occupies 2 bytes then the array has been stored in _____ order.	A	1	row major	column major	matix major	none of these
48	1	Consider a 2-D array x with 11 rows and 4 columns, with each element storing a value equivalent to the product of row number and column number. The array is stored in row-major format. If the first element x[0][0] occupies the memory location with address 1000 and each element occupies only one memory location, which all locations(in decimal) will be holding a value of 10?	B	1	1018, 1019	1022, 1041	1017, 1036	1000, 1399
49	1	A one dimensional array A has indices 1....75. Each element is a string and takes up three memory words. The array is stored at location 1120 decimal. The address of A[50] is	D	1	1264	1266	1164	1267

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50	1	Let A be a two dimensional array declared as follows: A : array [1... 10] [1... 15] of integer; Assuming that each integer takes one memory locations the array is stored in row-major order and the first element of the array is stored at location 100, what is the address of the element A[i] [j]?	A	1	15i+j+84	15j+i+84	10i+j+89	10j+i+89
51	1	For a two dimensional array A[-17….3][-3….6] the base address is 1020. The size of one element is 4B. What is the value of i and j for the 108th element if the arrays is stored is column major order?	C	1	-14,2	-15,1	-15,2	-14,3
52	1	Consider a Matrix A [-25....+25, -25....+25], base address (BA) = 0, size of element = 100 Bytes. Find the location of a [-20][-21] using RMO?		2				
53	1	Consider a Matrix A [-25....+25, -25....+25], base address (BA) = 0, size of element = 100 Bytes. Find the location of a [-25][-20] using CMO?		2				
54	1	Given an array, arr[1……10][1……15] with base value 100 and the size of each element is 1 Byte in memory. Find the address of arr[8][6] with the help of row-major order?		2				
55	1	Consider a two dimensional array A[20][10]. Assume storage size=4 bytes, the base address of array A is 100, elements are stored in row-major order and first element is A[0][0]. What is the address of A[11][5] ?		2				
56	1	The given array is P[-4…3][2…13] is a integer type array. An elements are stored in COLUMN MAJOR order. Calculate the address of 25th Element if base address is 100.		1				
57	1	Given array A [-4…3] [2…13] is an integer type array having base address 150. The elements are stored in column major order. Find the address of the 36th element of the array.		2				
58	1	The given array is A[4][4] is a floating type array. An elements are stored in COLUMN MAJOR order. Calculate the address of 11th Element if base address is 1020.		3				
59	1	Calculate the address of a[6][3] if the given array is a[9][10]with the base address 102 and floating type array. Find the answers of row major and column major representation respectively.		1				
60	1	Consider a 2-D array x with 11 rows and 4 columns, with each element storing a value equivalent to the product of row number and column number. The array is stored in column- major format. If the first element x[0][0] occupies the memory location with address 1010 and each element occupies only one memory location, which all locations(in decimal) will be holding a value of 8?		2				
61	2	Which of the following is an application of stack.	D	1	balancing of symbols	UNDO/REDO	recursion	All of these
62	2	In peep() pseudo code, which condition is to be called to check underflow	A	1	TOP – I + 1<=0	TOP – I + 1>=0	TOP – 1 + I<=0	TOP – 1 + I<=2
63	2	When stack is empty, which operation is not able to execute?	B	1	PUSH	POP	TOP = TOP + 1	TOP++
64	2	When data is to be deleted from data structure, but there is no element in stack, this situation is called ……	C	1	Overfull	Saturated	Underflow	Overflow

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65	2	Which one of the following is the process of inserting an element in the stack?	C	1	Insert	Add	Push	All of the above
66	2	When the user tries to delete the element from the empty stack then the condition is said to be a ____	B	1	Overflow	Underflow	Garbage collection	Global connection
67	2	If the size of the stack is 10 and we try to add the 11th element in the stack then the condition is known as___	A	1	Overflow	Underflow	Garbage collection	Global connection
68	2	Stack is a _____	A & C	1	LIFO	FIFO	FILO	LILO
69	2	Which function places an element on the stack?	B	1	pop	push	peep	isempty
70	2	_____ operation returns the value of ith element from the top of the stack.	C	1	push	pop	peep	change
71	2	Perform following test case on given stack of size 5. At the end of last operation, total number of elements present in the stack are – push(5); push(4); push(3); pop(); pop(); push(2); pop(); push(1);pop().	D	1	3	4	2	1
72	2	Which of the following is very useful Data structure when data have to stored and then retrieved in reverse order.	C	1	queue	LL	Stack	Tree
73	2	Adding element to stack means _____.	B	1	Placing element at the rear end	Placing element at the top	Placing element at the front end	both A or C
74	2	When new data are to be inserted into a data structure, but there is not available space, this situation is usually called	B	1	underflow	overflow	houseful	saturated
75	2	Inserting an item into the stack when stack is not full is called..... Operation and deletion of item from the stack, when stack is not empty is called ..... Operation.	A	1	push, pop	pop, push	peep, change	push, peep
76	2	The following operation performed on a stack of size 5. Push(1); Pop(); Push(2); Push(3); Pop(); Push(4); Pop(); Pop(); Push(5); After the completion of all operation, the total number of elements present in stack is?	A	1	1	2	3	4

77	2	Consider the following stack implementation.
		class Stack
		{
		static int size = 9;
		int arr[ ]=new int[size];
		int top = -1;
		}
		What would be the minimum value of the top such that the “overflow of the stack” message would be printed on screen?
			B	1	9	8	10	7

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78	2	Which statement is false for stack?	B	1	PUSH means insert one element in stack	POP means change position of element in stack	PEEP means search ith element in stack from top	CHANGE means change ith element in stack from top
79	2	TOP>=N means _____ and TOP=0 means _____ (As per Algorithm)	A	1	overflow, underflow	underflow, overflow	overflow, overflow	underflow, underflow
80	2	There are two stacks, stack1 and stack2. In the beginning both stacks are empty. Then we insert elements P, Q, R, S and T in same order in stack1. Now stack1 is popped 3 times and after each pop operation the popped element from stack1 is pushed into stack2. Now at this stage if I perform one pop operation on stack2 and print it on screen then what is the output on screen?	C	1	P	Q	R	S
81	2	Which of the following real-world scenarios would you associate with a stack data structure?	A	1	Piling up of chairs one above the other	People standing in a line to be serviced at a counter	Offer services based on the priority of the customer	Tatkal Ticket Booking in IRCTC
82	2	Match the following: 1. PUSH A. S[TOP-I+1] = X 2. CHANGE B. TOP+=1, S[TOP]=X 3. POP C. Return S[TOP-I+1] 4. PEEP D. TOP -=1, Return S[TOP+1]	A	1	1-B,2-A,3-D,4-C	1-A,2-B,3-D,4-C	1-B,2-A,3-C,4-D	1-C,2-A,3-D,4-B
83	2	The data structure required to check whether an expression contains a balanced parenthesis is?	A	1	Stack	Queue	Arrays	Trees
84	2	Which of the following conditions hold true when a stack is underflow on peep and the index of that array starts from 0.	D	1	Top-i+1<=0	Top>=N-1	Top-i+1=1	Top-i+1<=-1
85	2	User perform following operations on stack of size 5: push(1); pop(); push(2); push(3); pop(); push(2); pop(); pop(); push(4); pop(); pop(); push(5); then	B	1	Overflow Occurs	Underflow Occurs	Stack operations will occur smoothly	ArrayLimitsOutOfBounds Error is thrown
86	2	A stack is declared having a size of 10 elements, where the index of the stack starts from 0 and reaches upto 9 elements. If a stack has the following elements in it in the order bottom to top -> 11, 22, 33, 44, 55. If I want to change the value of the 2nd element from the top of the stack to element 99, then which of the following element value will be changed to element 99?	A	1	44	22	33	55

87	2	Consider the following pseudocode:
		declare a stack of characters
		while (there are more characters in the word to read)
		{
		read a character
		push the character on the stack
		}
		while (the stack is not empty)
		{
		write the stack's top character to the screen
		pop a character off the stack
		}
		What is written to the screen for the input "carpets"?
			C	1	serc	carpets	steprac	ccaarrppeettss

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88	2	The five items: A,B,C,D, and E are pushed in a stack, one after the other starting from A. The stack is popped four times and pushed back on the stack by popped order. Now one item is popped from the stack. The popped item is	B	1	A	B	C	D
89	2	The stack has the following order of elements inserted from bottom of stack to top of the stack -> 10, 20, 30, 40, 50. If the 2nd element from the top is to be peeped then which of the following is the correct option if the array index used to implement the stack starts from 1?	A	1	40	30	20	10
90	2	Which one of the following is not the application of the stack data structure	D	1	String reversal	Recursion	Backtracking	Asynchronous data transfer
91	2	Which data structure is mainly used for implementing the recursive algorithm?	B	1	Queue	Stack	Binary tree	Linked list
92	2	If the elements '1', '2', '3' and '4' are added in a stack, so what would be the order for the removal?	B	1	1234	4321	2134	2143
93	2	The minimum number of arrays required to implement a stack is __	A	1	1	3	2	5
94	2	If the sequence of operations – push(1), push(2), pop, push(1), push(2), pop, pop, pop, push(2), pop are performed on a stack, the sequence of popped out values are…	A	1	2, 2, 1, 1, 2	2, 2, 1, 2, 2	2, 1, 2, 2, 1	2, 1, 2, 2, 2
95	2	In CHANGE pseudo code “Procedure CHANGE(S, TOP, I, X) “, Where Stack is represented by S, TOP is a pointer, X is a new element to change and I is_____.	C	1	Vector	Array Pointer	Element’s Index from the TOP	Element’s Index from the BASE
96	2	Which of the following is not the correct statement for a stack data structure?	B	1	Arrays can be used to implement the stack	Stack follows FIFO	Elements are stored in a sequential manner	Top of the stack contains the last inserted element
97	2	Write a pseudo-code for PUSH and POP operations of stack.		6				
98	2	Write an algorithm for POP operation of Stack Data Structure		2				
99	2	Consider the stack S of characters, where S is allocated 8 memory cells. S: A,C,D, F, K, _, _, _ Describe the stack as the following operations take place. Pop(), Pop() ,Push(L), Push(P), Pop(), Push(R), Push (S), Pop()		3				
100	2	Write an algorithm for inserting an element and deleting an element in a stack.		7				
101	2	Write an algorithm for deleting an element from the stack.		3				
102	2	Write an algorithm for PEEP Operation and CHANGE Operation in a stack.		3				
103	2	List out different operations performed on stack and explain all with algorithm		7				
104	2	Write a Java program to implement a stack with all necessary overflow and underflow checks using array .		7				
105	2	Write a program to implement stack operations using an array (Operations: push, pop, peep, change, Display)		7				
106	2	Write an algorithm to reverse a string using stack.		5				

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107	2	Let S1 and S2 be two stacks. S1 has capacity of 4 elements. S2 has capacity of 2 elements. S1 already has 4 elements: 100,200,300 and 400, whereas S2 is empty, as shown below.Only the following three operations are available: Stack 1 - 400 (Top) 300 200 100 Stack 2 - PushToS2: Pop the top element from S1 and push it on S2. PushToS1: Pop the top element from S2 and push it on S1. GenerateOutput: Pop the top element from S1 and output it to the user. Note that the pop operation is not allowed on an empty stack and the push operation is not allowed on a full stack. Which of the following output sequences cannot be generated by using the above operations?	A	1	100, 200, 400, 300	200, 300, 400, 100	400, 200, 100, 300	300, 200, 400, 100
108	2	A function which calls itself is called _____	A	1	Recursion	Iteration	Algorithm	flowchart
109	2	Recursive function are implemented using _____	C	1	Library function	Recursion	Stack	None of these
110	2	Recursion occurs till _____	A	1	Stack overflow	Queue underflow	Linklist overflow	None of these
111	2	Recursion is not needed when you need to do which of the following operations?	A	1	Simply add two numbers 2 and 3	Find factorial of number 5	Find the GCD of 64 and 62	To solve 2 raised to 15
112	2	Recursion is similar to which of the following?	B	1	Switch Case	Loop	If-else	Array

113	2	Consider the following recursive function fun(x, y). What is the value of fun(4, 3)?
		int fun(int x, int y) {
		if (x == 0)
		return y;
		return fun(x - 1, x + y) ; }
			A	1	13	12	9	10

114	2	Which of the following recursive formula can be used to find the factorial of a number?	D	1	fact(n) = n * fact(n)	fact(n) = n * fact(n+1)	Both A&B	fact(n) = n * fact(n-1)
115	2	What does the following function do and how many times the function invokes for the value x=4 and y=3? int fun(int x, int y) { if (y == 0) return 1; return (x * fun (x, y-1)); }	C	1	It performs x+y and invokes 4 times	It performs x*y and invokes 4 times	It performs x^y and invokes 4 times	None of these

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116	2	In recursive Tower of Hanoi problem, for n number of discs, A is source, B auxiliary and C destination then complete the 3rd statement for the below function and also give the number for how many times the function invokes for the disc n=4?
		void TOH(int n,char source,char auxiliary,char destination){
		If(n>0){
		//Statement 1:
		TOH(n - 1, source, destination, auxiliary);
		//Statement 2:
		System.out.println(source + " -> " + destination);
		// Statement 3:
		?
		}
			A	1	TOH(n - 1, auxiliary, source, destination); & Function invokes 15 times	TOH(n, auxiliary, source, destination); & Function invokes 14 times	TOH(n - 1, source, auxiliary, destination); & Function invokes 15 times	None of these

117	2	What will be output for the following code?
		class Demo
		{ { public static void main(String args[]){
		System.out.println("Hello");
		Demo d = new Demo(); d.main();
		}
			B	1	Hello is printed once	Error	Hello is not printed at all	no output

118	2	Which of the following is the advantage of the recursion?	A	1	Recursion reduce the length of code.	Recursive functions are generally slower than non-recursive function.	It may require a lot of memory space to hold intermediate results on the system stacks.	Hard to analyze or understand the code.
119	2	When any function is called from main(), the memory is allocated to it on the stack.	A	1	TRUE	FALSE	Can be true or false	memory not required

120	2	What will be the output of this code?
		class ABC{
		int n = 1;
		void fun1() {
		if (n <= 20) {
		System.out.print(n+" ");
		n++;
		fun2(); } }
		void fun2() {
		if (n <= 20) {
		System.out.print(n+" ");
		n++;
		fun1(); } }
		public static void main(String args[]) {
		ABC a=new ABC();
		a.fun1(); }}
			A	1	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20	20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1	210	11 12 13 14 15 16 17 18 19 20

121	2	Consider the following recursive function.
		int function (int x,int y)
		{ if(y<=0)
		return x;
		return function(y,x%y);
		}
		The above recursive function computes_____.
			C	1	x*y	error	GCD of x and y	LCM of x and y

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D

122	2	Consider the following recursive function.
		void get(int n)
		{
		if(n<1) return;
		get(n-1);
		get(n-3);
		System.out.println(n);
		}
		if get(6) function is being called in main() then how many times will the get () function be invoked before returning to the main()?
			B	1	15	25	35	45

123	2	What does the following function do? int fun(int x, int y) { if (y == 0) return 0; return (x + fun(x, y-1)); }	C	1	x + y	x + x*y	x*y	x^y

124	2	What does the following function print for n = 25?
		void fun(int n){
		if (n == 0)
		return;
		System.out.print(n%2);
		fun(n/2);}
			B	1	11001	10011	11111	1010

125	2	What is the output of the following code?
		class Test {
		static int fun (int n) {
		if (n==4)
		return n;
		else
		return 2*fun(n+1); }
		public static void main(String args[]) {
		System.out.print(fun(2)); } }
			C	1	4	8	16	24

126	2	What is the output of the following code?
		class Test {
		static void fun(int n)
		{
		if(n == 0)
		return;
		System.out.print(n+ " ");
		fun(n-1);
		}
		public static void main(String args[])
		{
		fun(10);
		}}
			C	1	10	1	10 9 8 7 6 5 4 3 2 1	10 9 8 7 6

127	2	What is recursion? Write any program which follows recursion		5				
128	2	Write a program to find factorial of a given number using recursion		4				
129	2	Write a program to print fibonacci series for 10 numbers using recursion		4				
130	2	Write a program to print answer of x^y using recursion (Ex:- x=2, y=3, 2^3=8)		4				

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131	2	Write a program to calculate sum of digits of given number using recursion		4				
132	2	Write a program to find GCD of two numbers using recursion		4				

133	2	Consider the following pair of mutually recursive functions. What does g(g(2)) evaluate to?
		
		int f(int n){
		  if (n==0) return 0;
		  return f(n-1)+g(n-1);
		}
		int g(int n){
		  if (n==0) return 1;
		  return g(n-1) + f(n);
		}
			D	1	86	87	88	89

134	2	What will be the number of recursive calls for mystery(5) including the first call?
		
		void mystery(int n) {
		  if (n == 0 || n == 1) return 0;
		  mystery(n-2);
		  System.out.println(n);
		  mystery(n-1);
			A	1	15	31	63	127

135	2	Consider the following pseudo code based on standard stack operations :
		push(s,x) which pushesh element x on the top of the stack s.
		pop(s) which returns the top of the stack element after removing it from stack s.
		isEmpty(s) which returns true if stack s is empty and false otherwise.
		count_function(x):
		  s = empty
		  count = 1
		  for i = 2 to length(x):
		  if (x[i] == x[i-1]):
		  count++
		  else:
		  push(s, x[i-1])
		  push(s, count)
		  count = 1
		  end if
		  end for
		  push(s, x[length(x)])
		  push(s, count)
		
		  result = “”
			C	1	11	dcccccbbaaa	1d5c2b3a	d1c5b2a3

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		while not isempty(s):
		count = pop(s)
		char = pop(s)
		result = result + count + char
		end while
		return result
		What is the output for count_function("aaabbcccccd")?						

136	2	Write a java function to find HexaDecimal of a decimal number. Assume that Push(x) and Pop() functions are already available where Push(x) will push the element x into stack and pop() will return the top element by removing it from stack.		2				
137	3	Which of the following is the infix expression?	A	1	A+B*C	+A*BC	ABC+*	None of the above
138	3	The best data structure to check whether an arithmetic expression has balanced parentheses is ……	A	1	Stack	Linklist	Array	Queue
139	3	Which of the following is essential for converting an infix expression to the postfix form efficiently?	A	1	An operator stack	An operand stack	A or B	None of these
140	3	Consider the usual algorithm for determining whether a sequence of parentheses is balanced.The maximum number of parentheses that appear on the stack at any one time when the algorithm analyzes:(( )(( ))(( )))?	C	1	1	2	3	4
141	3	_____ is used to convert infix expression into postfix expression?	A	1	Stack	Linklist	Array	Queue
142	3	Which data structure is required to convert the infix to prefix notation?	B	1	Queue	Stack	Binary tree	Linked list
143	3	Which is the correct algorithmic sequence for the conversion of an expression from Infix to Prefix? A. Change of every '(' (opening bracket) by ')' (closing bracket) and vice-versa. B. Reversal of an infix expression. C. Conversion of the modified expression into postfix form. D. Reversal of postfix expression.	C	1	A, B, C, D	C, A, D, B	B ,A, C, D	D, B, A, C
144	3	The type of expression in which operator succeeds its operands is?	C	1	Infix Expression	Prefix Expression	Postfix Expression	Polish Notation Expression
145	3	Which of the following is the correct order of evaluation for the below expression? x+y%z/4*2−1	A	1	%/*+−	∗/%+−	/%+−*	∗/−+%
146	3	Which of the following is the correct order of evaluation for the below expression? x+y∗z/4%2−1	A	1	∗/%+−	∗/%+−	/%+−*	∗/−+%
147	3	What would be the polish notation for the given equation? (A*B) +(C+D)	A	1	.+*AB+CD	AB*CD*+	**AB+CD	AB+CD**
148	3	Which of the following is the prefix form of A+B*C?	D	1	A+(BC*)	+AB*C	ABC+*	+A*BC
149	3	Evaluate the prefix expression or polish expression × 5 – 4 3 using stack?		3				
150	3	Convert (A+B) *C-D^E^(F*G) infix expression into prefix format without using direct method.		3				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
151	3	Convert P $ Q $ R / S * T * U – V + W infix expression into prefix format		2				
152	3	Convert following Infix expression to Polish format showing stack status after every status after every step. A + (b / (B + a) ^ B * b) $ a * (A + B)		3				
153	3	Convert the infix notation into Polish notation. (you are not allowed to use direct method) A-B+(M$N)*(O+P)-Q/R^S*T+Z		3				
154	3	Convert the infix expression A × B + A × (B × D + C × E) into Polish notation using stack.		5				
155	3	Convert the infix expression (A + B) × (C + D) into Prefix or Polish notation?		5				
156	3	Convert the infix expression A × (B + C × (D + E)) / F × (G + H) into prefix notation?		5				
157	3	What would be the Prefix Expression for (S+T-M*N+(m^n)*w/u/V+t+Q) Infix Expression		2				
158	3	Evaluate the following infix expression. 2 $ 3 + 5 * 2 $ 2 – 6 / 6 (To solve this apply concept of evaluation of prefix expression).		1				
159	3	Which of the followings are valid RPN? I. abc*- II. abc+ III. ab+cd-* IV. abd/+-	D	1	I only	I, II only	I, II, III only	I, III
160	3	Assume that the operators +, −, × are left associative and ∧ is right associative. The order of precedence (from highest to lowest) is ∧, ×, +, −. The postfix expression corresponding to the infix expression a+b×c−d∧e∧f is	B	1	abc×+de∧f∧−	abc×+def∧∧−	ab+c×d−e∧f∧	− +a×bc∧∧def
161	3	What is the correct postfix expression for the following expression? a+b*(c^d-e)^(f+g*h)-i	C	1	abc^de-fg+*^*+i-	abcde^-fg*+*^h*+i-	abcd^e-fgh*+^*+i-	ab^-dc*+ef^gh*+i-
162	3	Infix To reverse polish conversion for given example: A+(B*C-D)+(E*(F/G-H))	B	1	ABCD*-+EFG/H+-*	ABC*D-+EFG/H-*+	AB*CD+-EFG/H-+*	ABCDEF/+-*GH-+*
163	3	What is the postfix expression for the corresponding infix expression? a+(b*(c*(d/e^f)*g)*h)	B	1	ab*cdef/^*g-h+	abcdef^/*g**h*+	abcd*^ed/g*-h*+	abc*de^fg/*-*h+
164	3	Convert 2 * 3 / (2 - 1) + 5 * 3 into Postfix form	B	1	2 3 * 2 1 - * 5 3 / +	2 3 * 2 1 - / 5 3 * +	2 3 * 2 1 + / 5 3 * -	2 3 / 2 1 - * 5 3 * +
165	3	What is the prefix expression corresponding to the expression? ((9+8)*7+(6*(5+4))*3)+2?	D	1	*++987**6++5432	*+++987**6_5432	=*++987**6+5432	++*+987**6+5432
166	3	Convert the infix notation (A – B + C × (D × E – F)) / G + H × K into postfix notation?		7				
167	3	Convert the following infix expression to postfix form using Stack. ((A – (B + C)) × D) / (E + F)		5				
168	3	Trace the conversion of infix to postfix form (A+B*C/D-E+F/G/(H+I))		5				
169	3	Convert following infix expressions to the postfix expressions. A/B$C+D*E/F-G+H		5				
170	3	Convert ((A – (B + C)) * D) î (E + F) infix expression to postfix form.		7				
171	3	Convert following infix expressions to the postfix expressions. A + B / C + D * (E - F) ^ G		7				

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Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
172	3	Convert the INFIX expression to POSTFIX expression and evaluate that POSTFIX expression by using following values. A+(B*(C-D))^ E $(F/G) A= 1, B= 2, C= 3, D= 1, E=2, F=1, G=2.		2				
173	3	Convert following infix expressions to the postfix expressions. (A+B) *D+E / (F+G*D) + C		3				
174	3	Convert the infix notation A × B + C × D + E × F into Postfix Notation or Reverse Polish Notation?		5				
175	3	Convert given expression from infix to postfix A / B ^ C + D * E - A * C		3				
176	3	Convert given expression from infix to postfix (6 +2) * 8/4		3				
177	3	Evaluation the following postfix expression by assuming J=1, K=2, L=3, M=5, N=5, O=2, P=7 and R=8 Expression: J+(K*(L*(M/N^O) *P) *R)		4				
178	3	Transform the following expression to postfix and evaluate the expression by assuming A=1, B=2, C=3, D=4, E=6, F=6, G=1, I=3, J=3 Expression: - A + B – C * D / E + F $ G / (I + J)		3				
179	3	convert 2*3/(2-1)+5*3 into Postfix form		5				
180	3	Construct the infix for / - A B * C ^ D E	C	1	A – B / C ^ D * E	A – B * C / D ^ E	(A – B) / (C * D ^ E)	A – (B * C ^ D) / E
181	3	Given the following prefix expression: * + 3 + 3 ↑ 3 + 3 3 3 What is the value of the prefix expression?	C	1	2178	2199	2205	2232
182	3	What would be the solution to the given prefix notation? * * + 1 2 / 4 2 + 3 5	D	1	40	8	50	48
183	3	What is the outcome of the prefix expression + - * 3 2 / 8 4 1?	C	1	12	11	5	4
184	3	What would be the solution to the given prefix notation? - * 1 5 / * / 6 3 6 2	C	1	1	0	-1	-2
185	3	Evaluate the Polish expression - + - x 4 3 2 50 / x 7 8 4 using stack?		7				
186	3	Transform the following postfix expression into its equivalent infix expression. A B C * D E F ^ / G * - H * +	C	1	(A + (((B*C) - ((D ^ (E/F)) * G)) * H))	(A + (((B*C) - ((D / (E^F)) / G)) * H))	(A + (((B*C) - ((D / (E^F)) * G)) * H))	(A * (((B+C) - ((D / (E^F)) * G)) * H))
187	3	The following postfix expression with single digit operands is evaluated using stack. 8 2 3 ^ / 2 3 * + 5 1 *- The top two elements of stack after the first * operator evaluated are.	A	1	6,1	5,7	3,2	1,5
188	3	The result evaluating the postfix expression 10 5 + 60 6 / * 8 - is	C	1	284	213	142	71
189	3	The equivalent infix expression and value for the postfix form 1 2 + 3 * 4 5 * – will be _____	A	1	((1 + 2) * 3) – (4 * 5) and -11	((2 + 1) * (3 – 4)) * 5 and 13	1 + 2 * (3 – 4) * 5 and -11	None
190	3	The result of postfix expression 5 3 * 9 + 6 / 8 4 / + is _____	B	1	8	6	10	9
191	3	What is the value of following suffix? A B + C D / * G H * + (where A=2, B=4, C=6, D=3, G=8, H=7)	D	1	40	280	140	68
192	3	What is the value of postfix expression 6324++*	D	1	Something between -5 and -15	Something between 5 and -5	Something between 5 and 15	Something between 15 and 100

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193	3	The equivalent prefix expression and value for the postfix form 10 2 + 24 8 / 2 * - will be _____	A	1	- + 10 2 * / 24 8 2 6	+ - 10 2 * / 24 8 2 -6	- + 10 2 / * 24 8 2 6	- + 2 10 * / 24 8 2 -6
194	3	Evaluate the following postfix expression using stack. 1 2 3 2 3 ^ ^ * + 6 2 / 4 * –		2				
195	3	Evaluate the following postfix expression using stack AB+CD/*GH*+ (where A=2,B=4,C=6,D=3,G=8,H=7)		4				
196	3	Evaluate the following postfix expression using a stack. Show the stack contents. AB*CD$-EF/G/+ A=5, B=2, C=3, D=2, E=8, F=2, G=2		5				
197	3	Evaluate: 623+-382/+*2^3+		2				
198	3	Evaluate the post fix expression or reverse polish expression 50 4 3 × 2 – + 7 8 × 4 / – using stack?		7				
199	3	Evaluate the following postfix expression. 7 6 + 4 * 4 10 + - 5 +		5				
200	3	The postfix for / + - a / b c e + a b	D	1	a b c / + e – a b + /	a b / c e – a b + + /	a b / c e - + a b + /	a b c / - e + a b + /
201	3	Convert the following Polish Expression to Reverse Polish Expression. Expression: *K$/L+-MNO^PQ		2				
202	3	1.Evaluate the following expression: a+(b*(c-d))^e$(f/g) where a=1, b=2, c=3, d=1, e=2, f=1 and g=2. 2. Convert the following expression from prefix to postfix: ++/a^bc/*d^efg^k$hj		2				
203	3	Write a program to evaluate the given prefix expression.		7				
204	3	Write a program to evaluate the given postfix expression.		7				